

CSE 220 Practice Problems: Continuous Fourier Transform & Fourier Series

Signal Processing Practice Bank

Category 1: 1D Fourier Series (Epicycles)

These problems involve modifying the 1D Fourier Epicycles framework.

Problem 1: The Time-Shift Property

Task: Modify the time array t to simulate starting the drawing from a different time offset, t_0 .

Theory: If $f(t) \leftrightarrow c_n$, then $f(t - t_0) \leftrightarrow c_n e^{-jn\omega t_0}$.

Implementation: Compute the coefficients for the original drawing ($c_{n,\text{orig}}$). Shift the input array z circularly to simulate a time delay, compute the new coefficients ($c_{n,\text{shifted}}$). Verify that $c_{n,\text{shifted}} = c_{n,\text{orig}} e^{-jn\omega t_0}$ by calculating the Mean Squared Error (MSE) of their magnitude and phase.

Problem 2: Parseval's Theorem for Fourier Series

Task: Prove that the average physical power of the drawing equals the sum of the squared magnitudes of the frequency epicycles.

Theory: $\frac{1}{T} \int_0^T |f(t)|^2 dt = \sum_{n=-N}^N |c_n|^2$

Implementation: Use numerical integration to integrate the absolute square of the spatial array over time, divided by T . Sum the squared magnitudes of your computed Fourier coefficients. Verify that the absolute difference between these two scalar values approaches zero.

Problem 3: Spatial Scaling and Rotation

Task: Apply a complex multiplier to the original drawing and prove the linearity of the coefficients.

Theory: Rotating the drawing by 45° and scaling it by 2 multiplies the spatial function by $Ae^{j\theta}$.

$Ae^{j\theta} f(t) \leftrightarrow Ae^{j\theta} c_n$

Implementation: Create a modified signal $z_{\text{mod}} = 2e^{j\pi/4} z$. Compute the Fourier coefficients of z_{mod} . Verify via MSE that they are exactly equal to your original c_n array multiplied by $2e^{j\pi/4}$.

Problem 4: Derivative (Velocity) of the Drawing

Task: Prove the differentiation property for the Fourier Series.

Theory: The velocity of the pen $\frac{d}{dt} f(t) \leftrightarrow jn\omega c_n$.

Implementation: Numerically differentiate the spatial array z with respect to t . Calculate the Fourier coefficients of this velocity array. Verify via MSE that they match your original c_n multiplied by $jn\omega$.

Category 2: 2D Continuous Fourier Transform

These problems involve modifying the 2D Image CFT framework.

Problem 5: 2D Spatial Shift Property

Task: Shift the image spatially and verify the 2D phase shift.

Theory: $\mathcal{F}\{I(x - x_0, y - y_0)\} = F(u, v) e^{-j2\pi(u x_0 + v y_0)}$

Implementation: Use circular rolling on the input image matrix I to shift the pixels. Compute the

2D CFT. Verify that the magnitude spectrum $|F(u, v)|$ remains identical. Verify the phase shift matches the theoretical exponential multiplier via MSE.

Problem 6: Low-Pass Filter (Blurring)

Task: Replace the high-pass filter with a low-pass filter to blur the image instead of finding edges.

Theory: A low-pass filter suppresses high spatial frequencies and preserves the components near the origin.

Implementation: Add a `low_pass(real, imag, cutoff)` method. For any $\sqrt{u^2 + v^2} > \text{cutoff}$, set the real and imaginary components to 0. Reconstruct the image and verify it produces a blurred version of the input.

Problem 7: 2D Parseval's Theorem (Energy Conservation)

Task: Verify that the total spatial energy of the image equals the total frequency energy.

Theory: $\int \int |I(x, y)|^2 dx dy = \int \int |F(u, v)|^2 du dv$

Implementation: Use 2D separable numerical integration on $|I(x, y)|^2$ over x and y . Do the same for $|F(u, v)|^2$ over u and v . Calculate the absolute numerical difference between the two total energy scalar values.

Problem 8: Edge Detection via Frequency Derivatives

Task: Compute the exact spatial gradients $\frac{\partial I}{\partial x}$ and $\frac{\partial I}{\partial y}$ using the frequency domain.

Theory: $\mathcal{F}\left\{\frac{\partial I}{\partial x}\right\} = j2\pi u F(u, v)$ and $\mathcal{F}\left\{\frac{\partial I}{\partial y}\right\} = j2\pi v F(u, v)$

Implementation: Multiply your complex spectrum $F(u, v)$ by $j2\pi u$ to get the horizontal gradient spectrum. Inverse transform it to get the physical horizontal edges. Do the same with v for vertical edges.

Problem 9: Conjugate Symmetry of Real Images

Task: Verify that because the input image $I(x, y)$ contains only real values, its frequency spectrum is perfectly symmetric.

Theory: $F(u, v) = F^*(-u, -v)$

Implementation: Extract the upper right quadrant of your computed $F(u, v)$ matrix. Extract the lower left quadrant. Apply a complex conjugate to the lower left quadrant and calculate the MSE between them to prove they are identical.

Problem 10: Gaussian Filter (Smooth Cutoff)

Task: Implement a Gaussian high-pass filter to mitigate Gibbs phenomenon (ringing artifacts) caused by hard cutoffs.

Theory: A Gaussian high-pass filter scales frequencies by $1 - e^{-\frac{u^2 + v^2}{2\sigma^2}}$.

Implementation: Add a `gaussian_high_pass` method. Multiply the frequency spectrum arrays by this exponential decay formula instead of using a binary cutoff radius. Reconstruct the image and compare.